Jonathan A. Klein

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EXPERIENCE

Senior Software Engineer, Cimpress, Remote **Jan 2026 – Present**

- Own core components of in-house manufacturing software, powering **$1.8B+** in annual revenue
- Led migration of a core event-tracking service (~5M events/day) from SQL Server to MongoDB Atlas, redesigning the data model for tenant-hierarchy queries and idempotent writes, and moving all historical data with 0 downtime
- Design and develop an in-house AI support pipeline using AWS Bedrock, automating workflows and reducing support requests by **50%**
- Deployed SQS and Lambda via infrastructure-as-code to handle high-throughput traffic, cutting request loss by **99%**

Software Development Engineer, Amazon, New York, NY **Jun 2025 – Jan 2026**

- Owned full-stack development and maintenance of ad management platform powering over **$3B** in annual revenue
- Launched self-serve revision APIs used by **150+** internal advertisers, reducing manual campaign updates by **70%** and cutting internal onboarding time by **40%**
- Built an internal chatbot using AWS Lambda and Bedrock to support team collaboration and employee onboarding
- Participated in a **24/7** on-call rotation, resolving production incidents and ensuring high availability of services

Head Teaching Assistant, Lehigh University, Bethlehem, PA **Jan 2024 – May 2025**

- Created assignments for Data Structures & Algorithms and Operating Systems courses (**350+** students total)
- Led recitation sessions and **8** office hours every week, providing personalized coding instruction and support
- Managed a team of **15** graders and coordinating the workflow for **30+** assignments

Researcher, Scalable Systems & Software Group, Bethlehem, PA **Dec 2022 – Nov 2024**

- Implemented CUDA-based GPU parallelization for zero-knowledge proof generation, achieving over **200%** speedup
- Optimized concurrent data structures in C++ by reducing lock contention and improving cache locality

Software Engineering Intern, FinFolio, Remote **May 2024 – Aug 2024**

- Refactored **50+** SQL queries and reviewed .NET code, improving financial report generation time by over **30%**
- Conducted root cause analysis to diagnose and resolve server-side bugs, enhancing system reliability

Data Science Intern, Mutual of America, New York, NY **Jun 2022 – Jul 2022**

- Implemented natural language processing (NLP) algorithms in Python to find correlations to client conversion rate
- Engineered pipelines to clean and process data, improving accuracy to over **90%** and reducing latency by **30%**

PROJECTS

Feature Flow: ML Feature Platform for Music Recommendation | GitHub

- Built an infrastructure-agnostic ML feature platform for music recommendation using Java 21 and Maven that enables both offline and online feature computation for ML training and inference
- Designed a stream processor serving multi-feature lookups over HTTP, reaching over **1M** requests per second
- Engineered fault-tolerant event processing with deduplication, recovery, and state management

Sockfish: UCI-Compliant Chess Engine | GitHub

- Engineered a low-latency chess engine using C++, featuring efficient board representation and move generation
- Accelerated graph search (iterative deepening, alpha-beta pruning), speeding up AI move selection by over **500%**
- Created a graphical user interface (GUI) using SFML for an intuitive user experience

SKILLS

- **Programming:** C++, TypeScript, Java, C#, SQL, Python, Rust, CUDA
- **Frameworks & Databases:** React, Angular, Node, .NET, MongoDB, Snowflake, MS SQL Server
- **Infra, DevOps, & Systems:** AWS (ECS, SQS, Lambda, Bedrock, S3, CloudFormation), Linux, Docker, Git

EDUCATION

Georgia Institute of Technology, Atlanta, GA **Expected Graduation: Aug 2028**
Master of Science in Computer Science, Concentration in Machine Learning **GPA: 4.0 / 4.0**
Lehigh University, Bethlehem, PA **Aug 2021 – May 2025**
Bachelor of Science in Computer Science & Mathematics **GPA: 3.96 / 4.0**

- **Relevant Courses:** Data Structures & Algorithms, Artificial Intelligence, Operating Systems, Parallel Computing, Databases, Advanced Algorithms, Programming Languages, Network Security, Graph Theory, Systems Software
- **Honors:** Summa Cum Laude, Upsilon Pi Epsilon, Tau Mu Lambda, Wilbur Mathematics Prize, Dean's List
- **Extracurriculars:** ACM (President), Mathematics & Statistics Society (Vice President), Chess Club